Subject Description Form

Subject Code	DSAI3201
Subject Title	Introduction to Natural Language Processing
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Knowledge in linear algebra, probability, and artificial intelligence is preferable.
Objectives	The objectives of this subject are to enable students to: 1. acquire solid understanding of fundamental concepts and techniques in NLP, including text processing, language models, and core tasks (e.g. parsing, classification, information extraction). 2. learn practical methods for building NLP systems using modern tools (e.g. Python libraries, machine learning frameworks) and apply them to solve problems. 3. develop critical thinking about language data and emerging NLP trends (e.g. large language models, agentic AI) and professional skills for teamwork and communication in NLP projects.
Intended Learning Outcomes *(Note 1)*	Upon completion of the subject, students will be able to: (a) understand the fundamental concepts, linguistic foundations, and major methodological approaches in natural language processing, including symbolic, statistical, and neural techniques; (b) apply core NLP techniques and machine learning models (e.g., text classification, sequence labelling, language modelling, and transformer-based models) to solve practical language processing tasks; (c) design, implement, and systematically evaluate an NLP system using appropriate tools, datasets, and performance metrics; (d) critically analyse recent advances, applications, and ethical implications of NLP technologies, including large language models and generative AI systems.
Subject Synopsis/ Indicative Syllabus *(Note 2)*	**1. Foundations of Natural Language Processing** Introduction to NLP concepts, history, and applications. Text preprocessing techniques (tokenization, segmentation, Unicode handling), corpus resources, and statistical language modelling (e.g., N-grams and smoothing). Fundamental linguistic analysis methods including part-of-speech tagging, morphology, syntax, and basic parsing. **2. Models and Techniques in NLP** Distributed representations and word embeddings (Word2Vec, GloVe, contextual embeddings). Neural architectures including RNNs (LSTM/GRU), sequence-to-sequence models, attention mechanisms, and transformer models (e.g., BERT, GPT). Applications such as text classification, sentiment analysis, machine translation, and information extraction.

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<tr><td></td><td>3. Applications of NLP
Dialogue systems, chatbots, and large language models (LLMs). Deployment considerations and evaluation strategies. Ethical and societal issues in NLP, including bias, fairness, transparency, and responsible AI practices.</td></tr>
<tr><td>Teaching/Learning Methodology
(Note 3)</td><td>Lectures introduce theory and concepts with examples; tutorials review key methods; hands-on labs give practical experience using NLP tools; a group project fosters deeper engagement. Students are encouraged to do independent reading of research and documentation. Emphasis is on experiential learning: building NLP pipelines and critiquing models in context. Active learning (in-class exercises, discussions of papers) supports critical thinking and teamwork.</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes
(Note 4)</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td></tr>
<tr><td>1. Tests/Quizzes</td><td>20%</td><td>✓</td><td></td><td></td><td>✓</td></tr>
<tr><td>2. Assignment/Project</td><td>20%</td><td></td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>3. Final Examination</td><td>60%</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr>
<tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr>
</table>
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Tests and quizzes assess students' understanding of fundamental NLP concepts, models, and techniques. They primarily assess by testing conceptual clarity, analytical ability, and continuous engagement with the subject content.

Assignments and the project assess students' ability to apply NLP algorithms and tools to real-world tasks, and to design, implement, and evaluate complete NLP solutions. This component primarily evaluates problem solving ability, while also fostering teamwork and technical communication skills.

The final examination provides a summative assessment of students' overall understanding of NLP principles and applications.</td></tr>
<tr><td>Student Study Effort Expected</td><td>
<table>
<tr><td>Class contact:</td><td></td></tr>
<tr><td>▪ Class activities (lecture, tutorial, lab, etc.)</td><td>39 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Assignments/projects</td><td>20 Hrs.</td></tr>
<tr><td>▪ Exams</td><td>30 Hrs.</td></tr>
<tr><td>▪ Self-study</td><td>33 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>122 Hrs.</td></tr>
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Reading List and References	**Reference Books:** 1. Jurafsky, D., & Martin, J. (2024). Speech and Language Processing (4th ed. draft). (Core Textbook); 2. Kamath et al., Deep Learning for NLP and Speech Recognition (2019); 3. Goldberg, Neural Network Methods for Natural Language Processing (2017); 4. Eisenstein, Natural Language Processing (2019); 5. Wolf et al., NLP with Transformers (2022); 6. Goodfellow Deep Learning (2016). 7. Online resources (e.g. NLTK book, PyTorch tutorials) will be recommended.